

Graph Neural Networks for Dynamic Spectrum Access

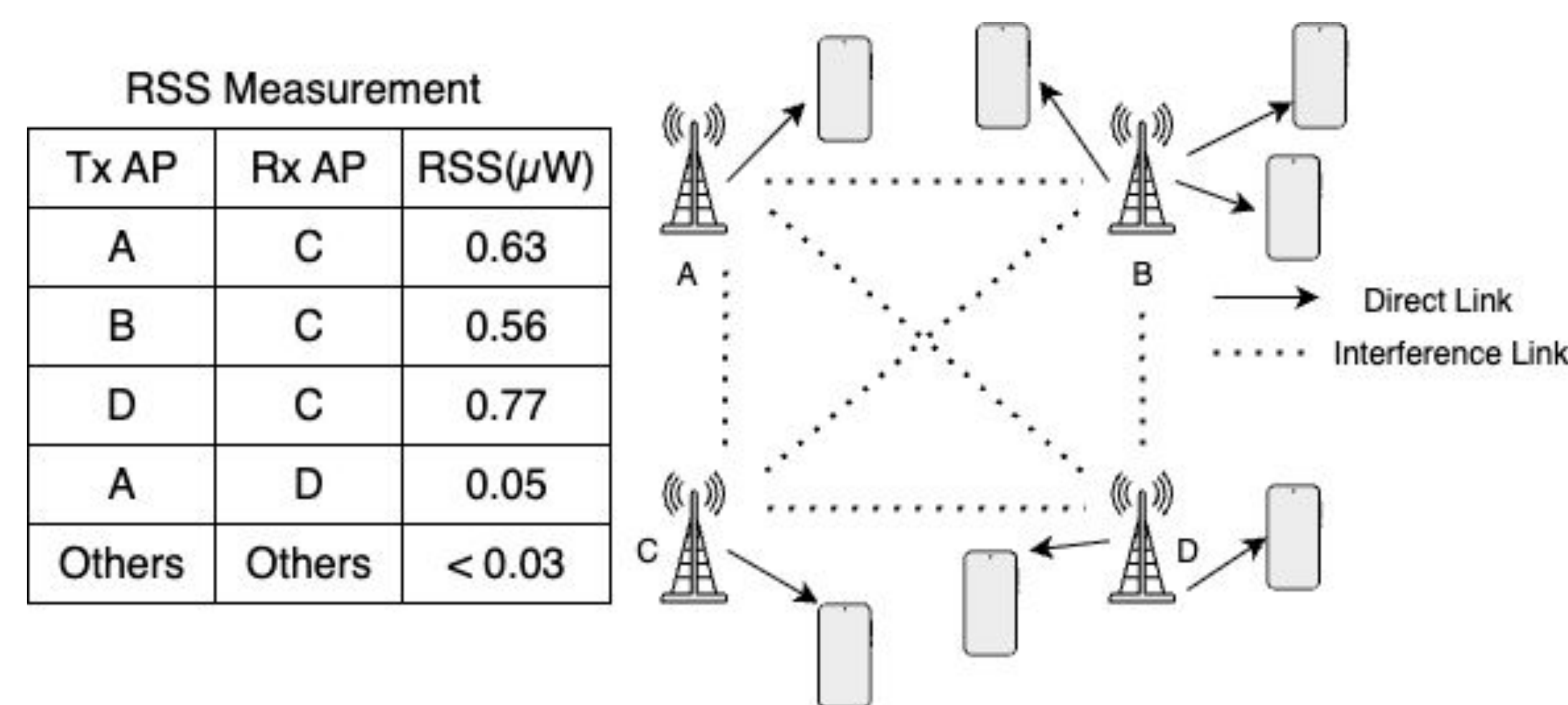
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SPECTRUM

BACKGROUND



- An efficient dynamic spectrum access framework requires two highly relevant components: Accurate knowledge of interference between access points (APs), and an allocation algorithm that uses this interference model to distribute spectrum efficiently.
- Pairwise interference can be measured using received signal strength (RSS), forming an $N \times N$ matrix that supports channel assignment and network planning.

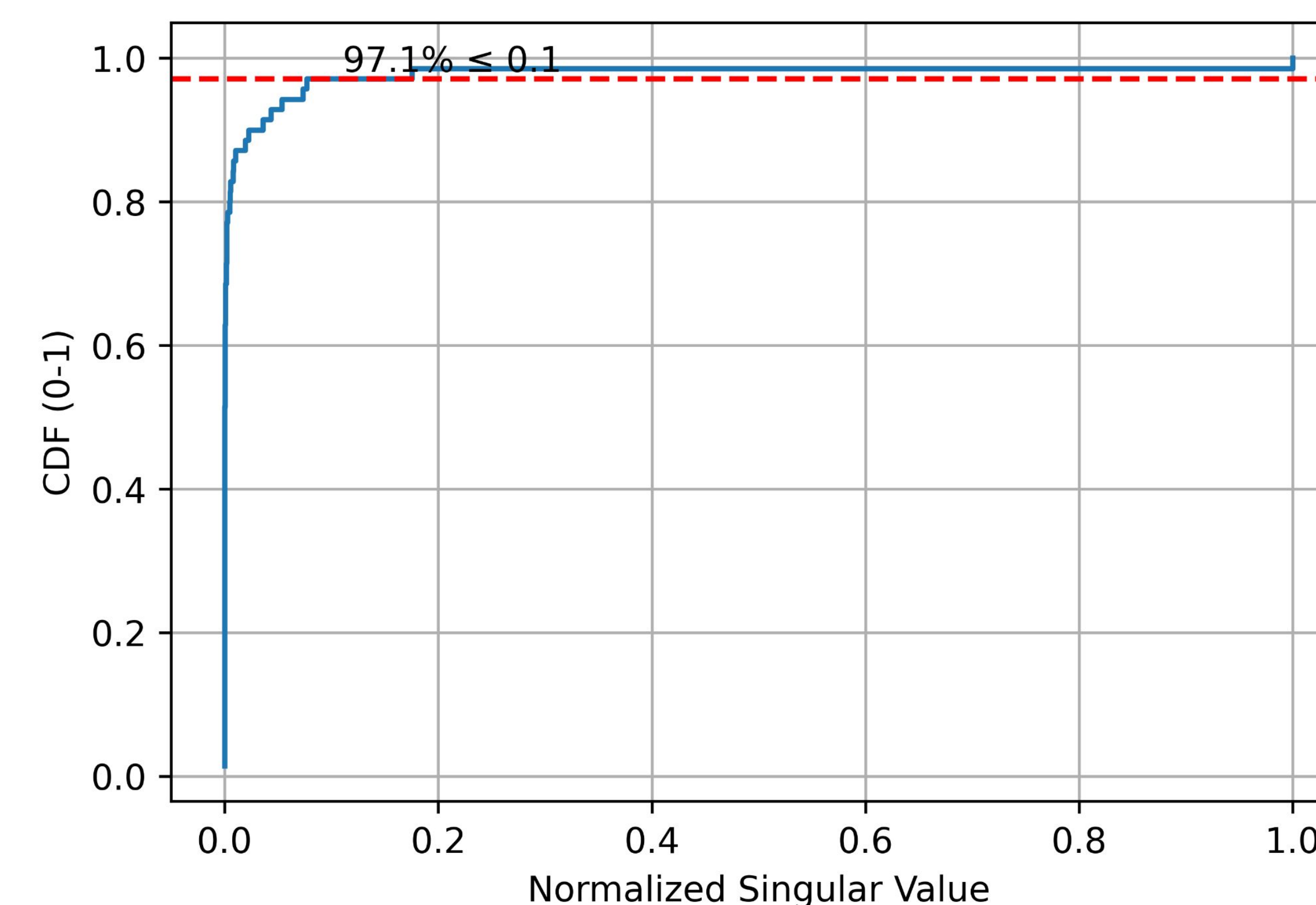
CHALLENGES

- Measuring all AP pairs costs $O(N^2)$ and is impractical at scale.
- Existing methods either require detailed 3D building maps, or per-site retraining, limiting their applicability in dynamic real world deployments.

RESEARCH QUESTIONS

- Can we accurately infer a complete AP-to-AP interference map from only a small fraction of pairwise RSS measurements, without requiring building maps or environmental models?
- Does a model trained once on a few environments generalize zero-shot to unseen deployments without retraining?

METHODS

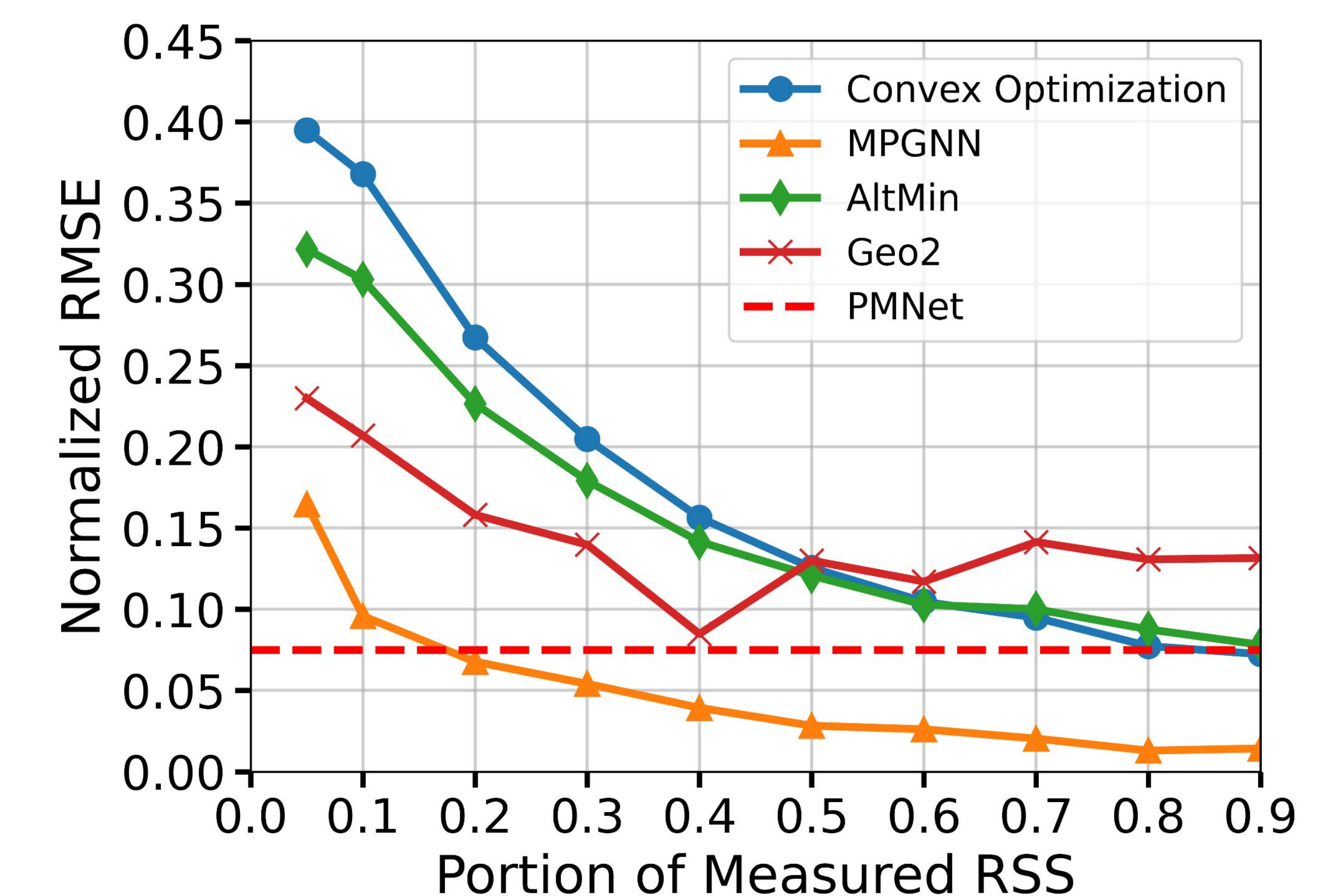


- By exploring real-world RSS datasets, we find that interference maps have low-rank structure and local geometric correlation. We leverage this structure to predict the complete interference map from sparse measurements.

We model each network as a complete directed graph: APs are nodes, AP coordinates are node features, and sparse RSS measurements are partially observed edge features.

- Input: AP coordinates + sparse RSS measurements
Output: Complete $N \times N$ RSS matrix – no building maps required.
- A two-layer MPGNN is trained with randomly masked observations, allowing it to generalize to arbitrary sparsity levels and apply zero-shot to unseen environments without retraining. Trained on 3 datasets and evaluated on 2 held-out environments

RESULTS



- Compared to matrix completion and ML-based baselines, MPGNN achieves the lowest prediction error, with the largest gains when only 5–10% of RSS entries are observed.
- NRMSE drops from 0.16 at 5% visibility to below 0.03 at 60% visibility. Up to 0.23 absolute NRMSE reduction over matrix completion baselines in the sparse regime
- Generalizes zero-shot to two unseen environments without retraining

INSIGHTS

- GNN is very robust against corrupted data measurement compared to optimization approaches.
- The execution time of GNN is efficient.
- GNN with Potts model as its loss function can effectively allocate spectrum.
- By separating the spectrum access problem into predicting interference map and allocation spectrum, we gain more interpretability and insight.